

Course and its outcomes

IV Semester: Computational Physics with Python (B011002T)

II: Elective – I

COMPUTATIONAL PHYSICS WITH PYTHON

- Unit I** Introduction to Operating systems, Linux, Python Basics: Interpreter, statements, variables, mathematical operators, loops, functions, libraries and modules, class and object, and simple applications.
- Unit II** Python libraries: NumPy, SciPy, matplotlib, pandas; extracting data from datafiles, managing datasets. 2D and 3D plots and graphs
- Unit III** Programming for Newton-Raphson method, iterative method, Newton's forward and backward interpolation.
- Unit IV** Numeric integration and differentiation, Euler's methods, Runge-Kutta methods for ODE, matrix manipulation
- Unit V** Programming for the motion of real pendulum, Random numbers, Monte-Carlo method, the random walk, Ising model and solution of wave equations, Schrödinger's equation.

Reading List:

1. Computational Physics with Python, Eric Ayars (California State University, 2013)
2. Computational Physics: Problem Solving with Python, 3rd edition, Rubin H. Landau, Manuel J. Paez, (Wiley-VCH, 2015)
3. Introducing Python: Modern Computing in Simple Packages, B. Lubanovic, (O'Reilly Media, Inc, 2015)
4. A Primer on Scientific Programming with Python, Hans Petter Langtangen (2014)

Learning Outcomes

Knowledge:

The students

- Will have fundamental understanding of different operating systems and working on Linux preferably Ubuntu.
- Will have knowledge of different features of Python programming language including module, package and libraries.
- Will learn about NumPy, SciPy, matplotlib and Pandas python libraries.

Skills:

The students

- Will be able to write computer programs for basic operation and problems.
- Will be able to manage and manipulate data in different data files for desired calculations and computation and to draw and analyze 2D and 3D plots and graphs for data sets and functions.
- Will be able to write computer programs for different numerical methods.
- Will be able to perform numeric integration and differentiation using computer program.
- Will be able to generate random numbers and solve waves equations including Schrödinger's equation using programming.